DEPARTMENT OF
SOFTWARE TECHNOLOGY

De La Salle University
College of Computer Studies
Department of Software Technology

Course Code	:	**CSALGCM**
Course Title	:	**Algorithms and Complexity**
Type of Course	:	Foundation course
Pre-requisites	:	CCDSALG (Data Structures and Algorithms)
Pre-requisite to	:	STALGCM (Algorithms and Complexity)
Term, AY	:	Term 1, AY 2026 - 2027

Instructor / Consultation hours / Contact details / Class details:

INSTRUCTOR DETAILS	SECTION	DAY	TIME	VENUE
Instructor: **Ryan Austin Fernandez** Contact via **Gmail (ryan.fernandez@dlsu.edu.ph)** Consultation Hours: **by appointment**	S03	T	0915-1045	ONLINE
		F	0915-1045	MM404
	S04	T	0915-1045	ONLINE
		F	1100-1230	MM404
Instructor: **Nathaniel Oco** Contact via **Gmail (nathaniel.oco@dlsu.edu.ph)** Consultation Hours: **by appointment**	S40A	M	1245-1415	V200A
		H	1245-1415	V200A
	S01	T	1430-1600	ONLINE
		F	1430-1600	A1110
	S02	T	1430-1600	ONLINE
		F	1615-1745	A1110

TYPE OF COURSE

☑ Hybrid	☐ Fully Asynchronous	☐ Predominantly Online
☐ Fully online	☐ In-person	☑ Predominantly In-Person

GENERAL COURSE DESCRIPTION

Computer scientists solve problems -- usually other people's problems. This course will provide insights into the foundational concepts in the art and science, primarily the latter, of problem-solving. An algorithm provides specifications to solve a "problem" on a "computer." In this course, we will look at the following topics:

- What exactly is a "problem"? Or a "computer"?
- If you are given two different algorithms to solve the same problem, how would you decide which is better? Or if you are given a single algorithm, how do you decide if it is "efficient" enough for you?

Several algorithmic design techniques apply to many problems. We will look at some of these :

- Divide and Conquer
- Dynamic Programming
- Greedy

Unfortunately, most practical problems are inherently intractable on modern-day computers. When you are given a problem, how would you recognize if it is intractable? To answer this, we look at the concept of NP-completeness.

TARGET SUSTAINABLE DEVELOPMENT GOALS (SDG)	
	**SDG 8 (Decent Work and Economic Growth):** Algorithm design and analysis can optimize systems and pipelines in common industries, increasing overall throughput and enabling economic growth in the sectors where these improvements are applied.
	**SDG 9 (Industry, Innovation, and Infrastructure)** This course aims to empower students with the foundational knowledge needed to build reliable applications by choosing the right data structures to manage information and determining the appropriate algorithms.
	**SDG11 (Sustainable Cities and Communities)** Algorithm design and analysis can optimize routing and urban planning, improve transportation, and reduce the carbon footprint of transit in these cities.

EXPECTED LASALLIAN GRADUATE ATTRIBUTES (ELGA)	
ELGA	**Course Learning Outcome**
CSCCT	Culturally Sensitive Critical and Creative Thinkers
EC	Effective Communicator
ESDC	Engaged and Service-Driven Citizen
TPP	Technically Proficient Professional

PROGRAM LEARNING OUTCOMES (PLO)	
Code	**Description**
CS01.02	Use appropriate computing knowledge within a specific application environment and domain knowledge in the design of computer-based solutions from defined requirements
CS01.03	Use appropriate computing knowledge within a specific application environment and domain knowledge in the implementation of computer-based solutions from defined requirements
CS01.04	Use appropriate computing knowledge within a specific application environment and domain knowledge in the assessment and evaluation of computer-based solutions from defined requirements
CS02.03	Model real-world problems as computational problems

CS04.01	Design solutions of varying levels of complexities, components or computing processes to meet the stated functional requirements
CS07.02	Write effective reports and documentations completely and comprehensively about the results of performing specific computing and professional tasks, including choices, assumptions and consequences of decisions
CS07.03	Deliver presentations effectively and efficiently to various audience (computing community, society at large, and users) using English and Filipino as needed, and respectful to the audience's background and culture

COURSE LEARNING OUTCOMES (CLO)	
Upon completion of this course, the student is expected to be able to do the following:	
CLO1	Design solutions using various problem-solving paradigms such as divide and conquer, greedy algorithms, and dynamic programming to solve various real-world problems.
CLO2	Investigate the running time and space consumption of algorithms designed through mathematical and empirical analysis.
CLO3	Identify whether problems are intractable and understand alternative solutions.

CLOs	SDG8	SDG9	SDG11
CLO1	IC	IC	IC
CLO2	IC	IC	IC
CLO3	IC	IC	IC

Code for the course contribution to the SDGs		Code Description
IC	Indirectly contributes	The course has minimal or indirect contribution to the achievement of the SDG because another SDG is more directly embedded in it or it is not intended to directly address any of the SDGs.
DC	Directly Contributes	The course directly contributes to the achievement of the SDG through its projects/activities.
PC	Potentially Contributes	The course is not intended to address any of the SDGs but its projects/activities has the potential to contribute to the SDG.

CLOs	CSCCT	EC	ESDC	TPP
CLO1	E	D	I	D
CLO2	E	D	I	D
CLO3	E		I	D

CLOs	CS01.02	CS01.03	CS01.04	CS02.03	CS04.01	CS07.02	CS07.03
CLO1	D			D	D		
CLO2			D			D	D
CLO3		E	E		E	E	E

Code for developmental levels (CLO-ELGA and CLO-PLO matrices)		Code Description Note: a PLO may have multiple codes if it addresses the outcome at different levels.
I	Introductory	Students are introduced to a new concept or skill.
E	Enabling	Enhancing, enriching, and improving what is introduced.
D	Demonstrative	Demonstrating the knowledge of skill for assessment

MAJOR COURSE OUTPUT

As evidence of attaining the above learning outcomes, the student is required to accomplish and submit the following outputs during the indicated dates of the term:

Learning Outcome	Required Output	Target Date
CLO1, CLO2	Hands-on Exam	Weeks 8 & 12
CLO1, CLO2, CLO3	Graded Quizzes	By instructor discretion
CLO1, CLO2, CLO3	Final exam	Week 14

RUBRIC FOR ASSESSMENT

Refer to the project specifications for each MCO.

OTHER REQUIREMENTS AND ASSESSMENTS

The course has the following requirements on top of the major course output described above:

1. Class Activities (recitation, short quizzes, etc.)
   - Information regarding the exact process of grading for this requirement will determine the exact grading process for this requirement in the first week of classes.
2. Assignments
   - Students are expected to complete assignments which may come in the form of problem sets, case studies, programming tasks, and/or reflection papers; these are means to refresh previous knowledge and/or encourage additional exploratory learning.
3. Long Exams or Short Quizzes
   - Students must take two written exams designed to test depth of their collective understanding of the course topics.
   - The schedule for the assessments will be determined by the teacher and students in due time during the term.

GRADING SYSTEM

To pass this course, one must accumulate at least 60 points through the course requirements discussed above.

Assessment	Maximum Points
2 Hands-on Exams	40
Final Exam	25
Quizzes	20
Class activities/assignments	15
Total Points	**100**

Minimum Points	GPE
94	4.0
89	3.5
83	3.0
78	2.5
72	2.0
66	1.5
60	1.0
Below 60	0.0

GENERATIVE AI USE POLICY

Assessment	Generative AI Use Policy	Remarks
2 Hands-On Exams	Banned	Exams are F2F. The only accessible resources are files placed in a drive folder beforehand. Lab moderation software will be used to restrict access to other sites.
Final Exam	Banned	The final exam is closed notes.
Quizzes	Banned	The quizzes are closed notes.
Class Activities / Assignments	Allowed in certain contexts	Generative AI may be used to help simplify or comprehend exercise items, but may not be used in design or analysis tasks.

This course is focused on developing competencies in the design and analysis of algorithms. While generative AI can augment comprehension and analysis of problems, there is also a risk of not being able to develop the core competency of algorithm design and analysis, which is a task that generative AI is not yet flexible enough to completely solve, considering the complexity of common problem spaces and the creativity required to develop appropriate, unique, and, at times, innovative solutions to real-world problems.

Therefore, in this course, students may not use generative AI in hands-on exams, final exams, or quizzes. Violations of this usage policy will be treated as academic dishonesty.

Learning Outcomes	Topics and Readings	Week/Date	Learning Activities
	Course Orientation Syllabus, Requirements, Class Policies	1	
CLO1, CLO2	**Computational Thinking** Problem Decomposition Abstraction Algorithmic Thinking Evaluation	1	Not limited to the following: ● Lecture ● Reading Assignments ● Homework / Writing Assignments
CLO1, CLO2	**Review of Algorithm Analysis** Introduction Growth of Functions Asymptotic Bounds (Big-Oh, Big-Omega, Big-Theta) Summations Recurrences Iteration Tree Method and Master Theorem	2	Not limited to the following: ● Lecture ● Short Quizzes/ Homeworks / Exercises ● Reading Assignments ● Programming Activities that can be bounded by a specified time complexity ● Perform empirical studies to validate hypotheses on runtime stemming from mathematical analysis ● Use recurrence relations to determine the time complexity of recursively defined algorithms and solve these recurrence relations
CLO1, CLO2	**Divide and Conquer (D & C)** Binary Search Bisection Method Merge Sort and Quick Sort Requirements and Limitations of Divide and Conquer	3 – 5	Not limited to the following: ● Lecture ● Reading Assignment ● Exercises &/or Programming activities where D & C is deemed appropriate
CLO1, CLO2, CLO3	**Dynamic Programming (DP)** Topics include, but are not limited to: Modeling DP Problems as a Directed Acyclic Graph (DAG) 0/1 Knapsack Coin Change Longest Increasing Subsequence Requirements and Limitations of Dynamic Programming *Note: Classes on October 5 and/or 6 will be asynchronous due to Faculty QA Training.*	6 – 8	Not limited to the following: ● Lecture ● Reading Assignment ● Exercises &/or Programming activities where DP is deemed appropriate Exercises &/or programming activities where one can identify practical examples or domains to which DP is deemed appropriate
Long Exam 1 – Wednesday/Thursday of Week 8			
Week 9: Independent Learning Week.			
CLO1, CLO2, CLO3	**Greedy Algorithms** Topics include, but are not limited to: Minimum Interval Cover	10 - 11	Not limited to the following: ● Lecture ● Reading Assignment

	Greedy Algorithm Problems involving Priority Queues Requirements and Limitations of Greedy Algorithms		Exercises &/or Programming activities dealing with Greedy methods
CLO3	**Introduction to Intractability** Topics include, but are not limited to: P, NP, EXP problems NP-hard and NP complete problems	12	Not limited to the following: ● Lecture ● Reading Assignment
Long Exam 2 – Wednesday/Thursday of Week 12			
CLO1, CLO2, CLO3	**Coping with hardness** Topics include, but are not limited to: Complete Search Recursive Backtracking	13	Not limited to the following: ● Lecture ● Reading Assignment ● Exercises &/or Programming activities dealing with backtracking and approximation methods ● Research Paper/Report on an approximation algorithm for an NP-hard/NP-complete problem
Final Exam			

TEACHING STRATEGIES

1. Lectures
2. Participative Discussions
3. Short Quizzes and Homeworks
4. Problem Sets
5. Programming Exercises
6. Reading Assignments
7. Project
8. Presentations
9. Written Assessments

REFERENCES

1. Cormen, T. H., Leiserson, C. E., & Rivest, R. L., Stein, C. (2022). Introduction to Algorithms. 4th Edition. MA: The MIT Press. **[Main Reference]**
2. Beecher, K. (2017, August). Computational Thinking. BCS.
3. Dasgupta, S., Papadimitriou, C. H., & Vazirani, U. V. (2008). Algorithms. McGraw-Hill Higher Education.
4. Goodrich, Michael T., Tamassia, Robert. (2011). Data Structures and Algorithms in Java. Wiley. [QA 76.73 J38 G66 2011]
5. Halim, S., Halim, F., Effendy, S. (2020) Competitive Programming 4. Lulu.
6. Kleinberg, J. & Tardos, E. (2006). Algorithm Design. Pearson/Addison-Wesley.
7. Sedgewick, R. (1989). Algorithms, 2nd Ed. MA: Addison-Wesley. [QA 9.58 S4 1989]
8. Skiena, Steven (2008). The Algorithm Design Manual. Springer. (To be Purchased)

9. Weiss, M. A. (1999). Data Structures and Algorithm Analysis in Java. MA: Addison Wesley. [QA 76.73 J38 W45 1999]

Software Tools:
GNU++15 C++ Compiler
JDK and JRE 8 Compiler and Runtime Environment for Java 1.8
Python 3 Programming Language

Web References for Software Tools:
Kattis (https://open.kattis.com)
Codeforces (http://codeforces.com)
HackerRank (http://hackerrank.com)
UVA Judge Online (http://onlinejudge.org/; http://uvatoolkit.com/)

CLASS POLICIES

- All university policies regarding attendance, cheating, the use of electronic devices, proper attire, eating, and other matters will be enforced. Refer to the Student Handbook for the details of these policies.
- In addition to these policies, the standard course policies enumerated below will also be enforced. Instructors may provide additional class policies.
- All requirements (assignments, project deliverables) are to be submitted during class time. Late submissions will be accepted only up to three (3) days after the deadline, with a 10% deduction from the highest possible score for each day late (rounded up to the nearest day).
- Missed class activities, such as cases, exercises, and short quizzes, cannot be made up.
- Students absent from the previous meeting(s) are expected to catch up on the missed discussion themselves (by asking their peers what they missed).
- Comply with Section 4.13 Policy on Academic Honesty of the DLSU Student Handbook. (Link: http://www.dlsu.edu.ph/offices/mco/publications/student_handbook/default.asp)
- Students are expected to email or text the instructor whenever they plan to skip a class session. Absences are excused if students inform the instructor within the day, but preferably before class time. Not doing so will be counted as an unexcused absence. Accumulated unexcused absences will result in a course failure.

DLSU PRIVACY NOTICE FOR ONLINE CLASS SESSIONS

- Online synchronous sessions may be recorded for future reference and shall be stored in **AnimoSpace** or a DLSU Google Drive account of the Faculty.
- Your name, profile photo, chat text, audio, and video data, which may contain education-related information, will be processed during the session.
- Do not share the meeting hyperlink or session information with those not intended to participate in the session.
- Do not personally record the session by means other than those authorized by the University, to respect the rights of every individual participating in the session.
- Do not re-share and/or repost the recording in other media or platforms to prevent unauthorized disclosure and processing of personal data.
- Discussions and conversations of a personal nature not related or relevant to the ongoing session are discouraged to prevent unintentional and/or unauthorized disclosure of personal data.

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